

Non-Functional Test Cases: Browser Compatibility Testing

Library Management System

Test Design Team (Non-Coders/Specification Experts)

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Note: These test cases are designed to verify the consistency of the application's appearance and functionality across different web browsers, based on SRS section 2.4 and the testing plan. The primary responsibility for execution lies with the Non-Coders/Specification Experts. Actual results and status are to be filled in during test execution.

IV.E Non-Functional Test Cases: Browser Compatibility Testing

Table 1: Browser Compatibility Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Testing Scope: Tests are to be executed on the latest stable versions of the following browsers as specified in the SRS and testing plan: Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.			
Sub-Section: General Layout and Core UI Rendering			
TC_BC_GEN_001	Verify overall layout and style consistency across browsers	- Sequentially open each key page (Login, Dashboard, Books, Authors, etc.) on all target browsers. - Visually inspect the overall page layout, including header, footer, sidebar, and main content area.	- The layout is consistent across all browsers. - All elements are correctly aligned. There are no overlapping or misaligned UI components. - Fonts, colors, and images render correctly and consistently. - No JavaScript errors are present in the developer console.
TC_BC_GEN_002	Verify responsiveness of the UI on different browser window sizes	- Open the application on a target browser. - Resize the browser window to simulate different screen sizes (desktop, tablet, mobile). - Observe how the UI elements reflow and adapt. - Repeat for all target browsers.	- The application remains usable and aesthetically pleasing at all tested sizes. - Navigation and content are accessible. - Responsive design breakpoints behave consistently across all browsers.
Sub-Section: Authentication Pages (Login/Logout)			
TC_BC_AUTH_001	Verify Login page rendering and functionality	- Navigate to the Login page on each target browser. - Inspect the rendering of input fields, labels, and the login button. - Enter valid credentials and click "Login".	- The login form elements are displayed correctly on all browsers. - User is able to type in the fields without issue. - Successful login redirects the user to the correct page, and this behavior is consistent across browsers.
TC_BC_AUTH_002	Verify error message display on login forms	- Navigate to the Login page on each target browser. - Attempt to log in with invalid credentials. - Observe the error message.	- Error messages are displayed clearly and styled correctly on all browsers. - The style and position of the feedback are consistent.
Sub-Section: Core Functionality (CRUD Operations)			
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Table 1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_BC_CRUD_001	Verify rendering of data tables/lists	• As a logged-in Librarian, navigate to a page with a data list (e.g., /books, /users). • Inspect the table's headers, rows, and pagination controls. • Repeat for all target browsers.	• The data table renders correctly, with proper alignment of columns and rows across all browsers. • Pagination controls are visible, functional, and look the same on all browsers.
TC_BC_CRUD_002	Verify functionality of forms and dialogs	• On a management page (e.g., /books), click the "New Book" button to open the creation form/modal. • Inspect all form fields (text inputs, dropdowns, checkboxes). • Trigger a confirmation dialog (e.g., by clicking a "Delete" button). • Repeat for all target browsers.	• Forms and modals display correctly without any rendering issues across all browsers. • All form controls (text fields, dropdowns, etc.) are fully functional. • Confirmation dialogs appear as expected and are functional on all browsers.
TC_BC_CRUD_003	Verify client-side form validation	• Open a creation form (e.g., "Add User"). • Submit the form with invalid data (e.g., incorrectly formatted email). • Observe the validation feedback. • Repeat for all target browsers.	• Client-side validation messages (e.g., "Invalid email format") appear correctly and consistently across all browsers. • The behavior (preventing form submission) is the same on all browsers.
Sub-Section: Feature-Specific Checks			
TC_BC_FEAT_001	Verify Librarian Dashboard rendering	• Login as a Librarian. • Navigate to the Dashboard (/dashboard). • Inspect the layout of statistics widgets and charts. • Repeat for all target browsers.	• All dashboard widgets and charts are displayed correctly. • The data visualization elements render properly and are legible on all target browsers. • There are no alignment or overlapping issues.
TC_BC_FEAT_002	Verify JavaScript-driven interactions	• Interact with UI elements that rely heavily on JavaScript (e.g., filtering a list, sorting a table, using a date picker in a form). • Perform these actions on each target browser.	• The interactive features work as expected on all browsers. • The performance and behavior of these scripts are consistent. • No browser-specific script errors are logged in the console.